

Enrico Caprioglio

Education

- 2022–present **PhD, Informatics**, *University of Sussex*, Brighton, UK.
Thesis submitted. Supervised by Luc Berthouze.
Thesis topic: mechanisms for integration and segregation in complex systems.
- 2020–2021: **MSc, Physics**, *Imperial College London*, London, UK, Degree classification: Distinction.
Dissertation topic: directed acyclic graphs embedded in spacetime.
- 2017–2020 : **BSc, Theoretical Physics**, *UCL*, London, UK, Degree classification: First-class honours.

Research Interests

Complexity science: frustrated systems; self-organization; emergence; synchronization; adaptive systems.

Neuroscience: whole-brain models; integration and segregation; metastability; brain criticality.

Computational social science: balance theory; polarization; homophily and heterophily.

Information-theory: multivariate systems; synergy; information decomposition; information dynamics.

Publications

Journal Articles (Published)

- 2026 **Enrico Caprioglio**, Pedro A. M. Mediano, and Luc Berthouze. Synergistic Motifs in Gaussian Systems. *Phys. Rev. Lett.*, volume 136, page 127401. American Physical Society, Mar 2026. doi:[10.1103/179y-qp6j](https://doi.org/10.1103/179y-qp6j).
- 2024 **Enrico Caprioglio** and Luc Berthouze. Emergence of metastability in frustrated oscillatory networks: the key role of hierarchical modularity. *Frontiers in Network Physiology*, volume 4, 2024. doi:[10.3389/fnetp.2024.1436046](https://doi.org/10.3389/fnetp.2024.1436046).

Preprints and Manuscripts Under Review

- 2026 **Enrico Caprioglio**, and Luc Berthouze. Heterophily as a generative mechanism for self-organized synergistic interdependencies, [arXiv:2604.11545](https://arxiv.org/abs/2604.11545) [physics.soc-ph].

Books and Chapters

- 2020 Mario Campanelli, Antonio d’Alfonso Del Sordo, Camilla Tacconis, **Enrico Caprioglio**, Lodovico Scarpa, Muhammad Tayyab Shabbir, and Sheila María Pérez García. *Classical Mechanics: A professor–student collaboration*. IOP Publishing, 2020.

Conferences, workshops and schools

Selected Talks and Conference Presentations

- May 2026 **BritishNetSCI26**, *London, UK*.
Contributed talk: “Heterophily as a generative mechanism for self-organized synergistic interdependencies”.
- June 2025 **NetSci**, *Maastricht, Netherlands*.
Contributed talk: “[Synergistic Motifs in Linear Gaussian Systems](#)”. Presented at [TopoNets satellite](#).

- July 2024–2026 **London International Youth Science Forum, London, UK.**
Invited talks: “Self-organized criticality: the study of self-organization” and “Complexity science: the study of emergence”.
- June 2024 **NetSci, Quebec City, Canada.**
Contributed talk: “Emergence of metastability in frustrated oscillatory networks: the key role of hierarchical modularity”. Presented at [Network Neuroscience satellite](#).
- April 2024 **CompleNet, Exeter, UK.**
Contributed talk: “Competition across timescales: emergence of metastability in large networks of oscillators”.
- January 2022 **Wolfram Winter School, remote.**
Project titled: “An investigation on possible notions of physical distance in the WM”. [Available at wolfram.community](#).
[Workshop and School Organization](#)
- October 2025 **Workshop on Emergence, University of Sussex, Brighton, UK.**
- October–December 2019 **Physics Taster Course, University College London, London, UK.**
Organized and taught in a five-week physics taster course based on the co-authored textbook: Classical Mechanics, a professor student collaboration.

Supervision

- Summer 2024: **MSc dissertation supervision, Department of Informatics, University of Sussex.**
Supervised two student dissertation projects. Provided guidance on research design, literature review, mathematical/computational methods, and final dissertation preparation.

Teaching Assistantships

- Spring 2025 and 2026 **981G5: Network Science, University of Sussex, Brighton, UK.**
Also delivered lecture 6 (2 hours) on stochastic block models and Bayesian inference.
- Autumn 2022 and 2023: **817G5: Mathematics and Computational Methods for Complex Systems, University of Sussex, Brighton, UK.**
Including preparation of preliminary materials and extra readings.

Professional / Outreach Experience

- August 2017 – August 2023 **Hall Host, London International Youth Science Forum (LIYSF).**
Organization of large events (500 students from 90 countries) which included talks from Nobel prize winners and British royal family members. Position of responsibility; chairing lectures; organization of talks, lectures and visits across the UK.

Honours and Awards

- July 2022 **Best project award, Mediterranean School of Complex Network, Online Participant.**
- May 2017 **Honorable Mention, ISWEEEP, Houston, US.**
- January 2017 **Innovation Prize, Youth Category, LEONARDO 2016, Rome, Italy.**
- March 2016 **Project Finalist, FAST, I Giovani e le Scienze (Science and Youth), Milan, Italy.**

Miscellaneous

- Technical Expertise Python; Julia; Mathematica; LaTeX; Git; network science methods; dynamical systems theory; stochastic block models; information theory; Monte Carlo methods; data analysis.
- Peer review Frontiers in Physiology and UNESCO
- Internal seminars: Centre for Complexity Science, Imperial College London, UK, 2025; MPIE conference, 2024–2025; Sussex Neuroscience, 2024.